



The effects of employers' disability and unemployment insurance costs on benefit inflows[☆]

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ABSTRACT

In Finland, large firms are partially liable for the costs of disability and unemployment benefits paid to their former workers. To estimate the effects of such cost liabilities, we exploit a reform that extended this cost-sharing to cover a new group of blue-collar workers. We show that experience rating in disability insurance reduces inflows to sickness and disability benefits and increases participation in vocational rehabilitation programs, whereas employers' unemployment insurance costs reduce excess layoffs of older workers who can qualify for extended unemployment benefits until retirement age. We find no evidence of spillover effects: employers' cost liabilities in one benefit type do not affect inflows to other types of the benefits.

1. Introduction

Disability and unemployment benefit schemes are among the largest social security programs in many countries. In 2019, public spending on incapacity benefits (i.e. sickness and disability benefits) amounted to 2% and unemployment expenditures 0.6% of GDP in the OECD countries. A large part of disability insurance (DI) and unemployment insurance (UI) costs is financed through employer contributions. In most countries, firms pay flat-rate DI and UI premiums, which may depend on the age structure of the workforce or industry but not directly on the costs of the benefits received by the firm's own employees. This can lead to inefficiently high benefit inflows because firms do not need to take into account the societal costs of new benefit claims when making decisions about layoffs and investments in workplace health and safety. The government may aim to mitigate this incentive problem through experience rating or coinsurance. With experience rating, the firm's insurance premium rate (i.e. payroll tax rate) increases with the amount of benefits awarded to its employees in some past reference period. With coinsurance, the firm has to pay a lump sum contribution

to the insurance provider at the time when the benefit is awarded to its employee. In the both cases, the firm is directly liable for part of the benefit costs of its own employees. The objective is to make employers internalize the costs to society of new benefit claims.

Coinurance and experience rating systems are controversial because the administrative costs can be high while the evidence of their effectiveness in reducing benefit costs is limited and mixed. Such systems may also distort hiring decisions and have undesired spillover effects on the receipt of other types of benefits. Perhaps this uncertainty explains why only a few countries have adopted an experience rating or coinsurance system to finance their UI and DI expenditures. The United States is the only country where firms pay experience-rated UI premiums,¹ whereas it is only in Finland and the Netherlands that firms pay experience-rated DI premiums. In some European countries, including Finland, France, Italy and Germany, firms must pay lump sum contributions for layoffs in certain cases (Fath and Clemens, 2005). Evidence on the effects of experience rating and coinsurance should not only be of interest for policy makers in these countries but also for those

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¹ It is not clear why the United States and European countries have adopted different approaches to finance their UI systems despite other similarities in the design of UI. One possibility is that more severe employment protection laws in Europe act as a substitute for the experience-rated UI premiums in the United States (Fath and Fuest, 2005). This is because both employment protection (e.g. mandatory severance payments) and experience rating increase the costs of laying off workers, and can be used to reduce inefficiently high labor turnover.

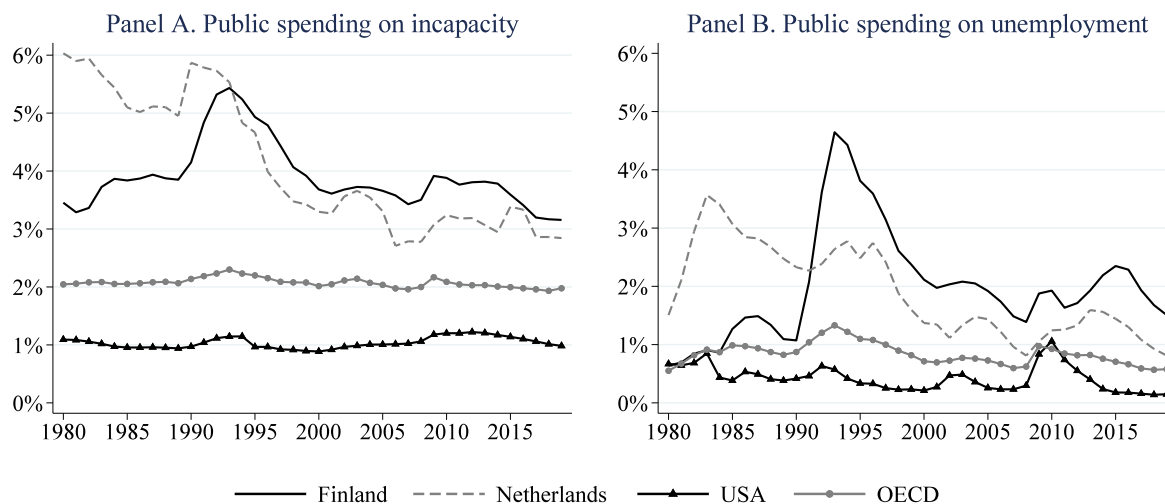


Fig. 1. Public spending on incapacity and unemployment in 1980–2019, % of GDP. Note: Public spending on incapacity refers to spending due to sickness, disability and occupational injury. Public unemployment spending is defined as expenditure on cash benefits for people to compensate for unemployment. Source: OECD's Social Expenditure Database.

in other countries who are considering alternative reforms to cut the DI and UI expenditures. Autor (2011) and Burkhauser and Daly (2011), for example, have suggested that the US Social Security DI program should be financed by an experience-rated payroll tax.

In this paper, using data from Finland, we show that experience rating and coinsurance can be useful policy tools to reduce disability and unemployment inflows. Finland provides a particularly interesting case for studying the effects of such policies because the widespread use of unemployment and disability benefits as early retirement pathways has been a serious problem in Finland for a long time. This problem can be attributed to certain institutional characteristics that facilitate early retirement, such as generous unemployment benefits and looser eligibility criteria for disability benefits applied to older workers. Finland's public spending on incapacity and unemployment benefits peaked in the early 1990s during a deep recession when the unemployment rate rose to above 16% (Fig. 1). Since the mid-1990s, a series of reforms to promote longer working careers has been implemented. As a result, incapacity and unemployment expenditures have been trending down, but they have remained above the OECD averages.

In Finland, large firms pay DI premiums that depend on the past disability benefit claims of their former employees. Large firms are also partially liable for the costs of extended unemployment benefits paid to their former employees. Extended benefits are available for older unemployed individuals who exhaust their regular UI benefits. When a benefit extension is awarded, the former employer is obligated to pay a share of the estimated costs of the extended benefits over the coming years as a lump sum payment to the Unemployment Insurance Fund. Small firms are not subject to experience rating in DI, nor coinsurance in UI.

In 2007, a pension reform extended the coverage of experience rating and coinsurance to a new group of blue-collar workers in construction, dock work, agriculture and forestry. As a consequence, large employers became directly liable for the DI and UI costs of these workers, whereas smaller employers were not affected. This worker group is of particular interest due to its exceptionally high disability and unemployment risks. We exploit the 2007 reform to identify the causal effects of experience rating in DI and coinsurance in UI using a difference-in-differences (DID) approach.

We find that experience rating in DI reduces inflows to sickness and disability benefits, and increases participation in vocational rehabilitation programs. Like previous studies (e.g. Hakola and Uusitalo, 2005, Kyrrä and Wilke, 2007, and Kyrrä and Pesola, 2020), we find a higher layoff risk for older workers who can qualify for extended UI benefits. The layoff risk of this group is particularly high in large firms,

suggesting that large employers may exploit the extended UI benefit scheme as a soft way to reduce employment when downsizing. We show that coinsurance in UI alleviates this problem by reducing excess layoffs of the oldest workers. We find no evidence of adverse spillover effects: experience rating in DI does not increase layoffs, nor does coinsurance in UI increase the inflow to disability benefits. We also show that the desired effects of experience rating on benefit inflows are not driven by employers becoming more selective when hiring new workers.

Our study contributes to the literature on the effects of employer liabilities in DI and UI. A handful of papers have analyzed the Dutch DI system, providing somewhat mixed results.² The findings of Koning (2009) and van Sonsbeek and Gradus (2013) suggest that the introduction of experience rating in 1998 reduced the inflow to disability benefits in the Netherlands.³ de Groot and Koning (2016) show that the removal of experience rating from small firms increased the disability inflow by 7% over the years 2003–2004, but they find no effect when experience rating was re-introduced in 2008. The asymmetry in the effects may be due to other changes in the sickness and disability schemes between the two reforms.⁴ The introduction of experience rating in the late 1990s was part of a series of reforms in disability and sickness programs that started in the early 1990s and continued over the next two decades. These reforms were a response to an exceptionally high level of DI recipients, which was considered to be one of the major social and economic problems of the Netherlands (Koning

² Several authors have analyzed experience rating in US and Canadian Workers' Compensation programs, which cover medical costs and provide cash payments for those who are injured at work (e.g. Ruser, 1985, 1991; Bruce and Atkins, 1993; Krueger, 1990; Thomason and Pozzebun, 2002). These studies present evidence that experience rating reduces on-the-job injuries and the duration of injury spells. Because employers have less control over disability outcomes than workplace injuries, we should expect smaller effects for experience rating in DI.

³ de Groot and Koning (2022) find that higher experience-rated DI premiums also increase the probability of bankruptcy among small Dutch firms. This is unlikely to be a problem in Finland where only relatively large firms are subject to experience-rating and coinsurance costs.

⁴ In 2005, the period of sickness benefits was extended from one to two years. This extended employers' liabilities, given that Dutch firms are responsible for the costs of sickness benefits for their employees. One year later, separate disability benefits were introduced for persons who are permanently fully disabled and for persons who are only partially and/or temporarily disabled. Since this reform, experience-rated DI premiums have been applied only to the disability benefit costs of the latter group.

and Lindeboom, 2015). Following the reforms, public spending on incapacity benefits as a percentage of GDP halved from 6% in the early 1990 to 3% in 2005 (Fig. 1). According to Koning and Lindeboom (2015), the most important lesson from the Dutch experiences is that employers should be encouraged to seek ways to prevent long-term sickness and absence, and subsequent disability inflow.

In Finland, a coinsurance system in DI existed for several decades until it was replaced with an experience-rating system in 2006, when the Finnish firms began to follow the International Financial Reporting Standards in their financial statements, as required by the European Union. Because the coinsurance system was hard to integrate in the new accounting environment, it was abolished and replaced with experience-rated DI premiums. This change was made purely for technical reasons, and the experience-rating system was designed to mimic the incentive structure of the old coinsurance system. Korkeamäki and Kyrrä (2012) and Hawkins and Simola (2018) have studied the incentive effects of the old system. The former study finds that coinsurance in DI reduced the inflow to sickness benefits and further transitions from sickness benefits to disability benefits. The latter provides evidence that a higher degree of coinsurance discouraged employers from hiring workers with a high disability risk.

Kyrrä and Paukkeri (2018) study the effects of the current experience-rating system among worker groups that were not affected by the 2007 pension reform. These worker groups have lower disability and unemployment risks than the group studied in this paper. Kyrrä and Paukkeri (2018) find that experience rating has little or no effect on inflows to sickness and disability benefits, suggesting that replacing coinsurance with the experience-rating system in 2006 may have weakened the incentive effects and possibly increased overall DI costs.⁵ We complement this research by showing that experience rating reduces sickness and disability benefit inflows at least in certain blue-collar occupations with exceptionally high disability rates, and that experience rating increases participation in vocational rehabilitation programs. Also, it should be stressed that Kyrrä and Paukkeri (2018) cannot rule out moderate-sized effects of experience rating due to the limited statistical power of their regression-kink setting.⁶ In this paper, we provide more convincing causal evidence based on a difference-in-differences approach.

Very little is known about the effects of the experience rating of UI premiums in the US labor market due to the lack of appropriate data and exogenous variation in the premium rates. Anderson and Meyer (2000) and Woodbury et al. (2004) provide some empirical evidence that a higher degree of experience rating in UI reduces layoffs, while earlier studies such as Topel (1983) and Card and Levine (1994) have found that experience rating reduces temporary layoffs. A recent study by Johnston (2021) finds that an increase in the firm's UI premium due to layoffs in the past three years reduces hiring and employment but does not affect layoffs. We complement this literature by providing causal evidence on the effects of coinsurance in UI.⁷

Our paper is also related to the literature on spillover effects between social benefit programs. Several authors have found evidence of

substitution between DI and UI schemes in the Netherlands. Koning and van Vuuren (2010) find that a certain part of desired layoffs took place through the DI program but find no evidence that disabled workers would have ended in UI benefits instead of DI benefits. Borghans et al. (2014) show that DI recipients responded to a reduction in their DI benefits by collecting more benefits from other social assistance programs, while Lammers et al. (2013) find that an increase in the stringency of job search requirements encouraged older unemployed individuals to claim DI benefits. Karlström et al. (2008) and Staubli (2011) analyze the effects of reforms that tightened eligibility criteria for DI benefits among older workers in Sweden and Austria, respectively. Both studies find higher exit rates to unemployment and sickness benefits following the reforms. Kyrrä (2015) analyzes a similar DI reform in Finland but finds no effect on the unemployment inflow. Inderbitzin et al. (2016) find that availability of extended UI benefits reduces take-up of DI benefits among older workers in Austria, whereas Kyrrä (2015) and Kyrrä and Pesola (2020) do not find that evidence that eligibility for extended UI benefits would affect the inflow rates to disability and sickness benefits in Finland.⁸ Overall, the previous studies have found no spillover effects between UI and DI programs in Finland, which is at odds with the evidence from the Netherlands, Sweden and Austria. Identification of the spillover effects in these studies is based on changes in eligibility rules or benefit generosity (Koning and van Vuuren, 2010 is an exception), so that the estimated effects may be mainly driven by the labor supply responses of affected individuals. The present study complements this literature by studying potential spillover effects between unemployment and disability programs that may arise from employer behavior.

The remainder of the paper proceeds as follows. Section 2 describes the institutional setting. Section 3 discusses how the 2007 reform changed employer liabilities for certain worker groups and employers. Section 4 describes the data used and Section 5 reports sample statistics. Section 6 discusses the econometric methods and reports the results. Section 7 concludes.

2. Institutional framework

2.1. Benefit schemes

A worker who cannot work due to a medical condition is entitled to compensation for income loss. For the first weeks (typically one to three months depending on the collective agreement), the applicant is fully compensated by the employer, after which he or she may receive a *sickness benefit* up to about one year. The sickness benefit is paid by the Social Insurance Institution. During the sickness benefit period, the worker's health status and rehabilitation needs are repeatedly evaluated in co-ordination with the employer, occupational health provider and Social Insurance Institution.

A worker who is at risk of becoming disabled may apply for vocational rehabilitation (work trial, job coaching, apprenticeship training or vocational retraining program). The objective of vocational rehabilitation is to prevent disability and improve employment possibilities when a worker can no longer continue at his or her current job for health reasons. The health problem that hinders the worker from performing his or her current job tasks must be verified by a physician. For the duration of vocational rehabilitation, the worker receives either pay from their employer or a *rehabilitation allowance* from the pension institution. In the former case, the rehabilitation allowance is paid to the employer. Since 2004, occupational rehabilitation has been a subjective legal right for employed workers who are at risk of work-capacity loss. Occupational rehabilitation programs are arranged and

⁵ Their estimates imply no significant effect on average in the population of all workers. However, they find a significant effect on the disability inflow for men under the age of 50, but not for other groups.

⁶ Their identification strategy was based on kinks in the formula that determines the degree of experience rating in DI as a function of firm size. Since these kinks are quite small, the regression-kink approach produced rather imprecise estimates.

⁷ Prior to the current cost-sharing in UI in Finland, the long-term unemployed were entitled to an unemployment pension at age 60, and the former employer had to pay a lump sum coinsurance contribution to the pension provider when the unemployment pension was awarded. Hakola and Uusitalo (2005) show that higher unemployment pension costs reduced layoffs of older workers. The unemployment pension scheme was abolished in 2005, after which large firms have been liable for the costs of extended UI benefits paid to their former employees.

⁸ Using more recent data for Finland, Yashiro et al. (2022) find some evidence that eligibility for extended UI benefits may have a positive effect on the DI inflow. However, this result was not robust with respect to the model specification.

financed by the pension institutions in co-ordination with employers. About two-thirds of the participants of occupational rehabilitation programs return to employment.

In case of prolonged disability, the individual may qualify either for a *disability pension* or *rehabilitation benefit*.⁹ If it is unlikely that the applicant will be able to return to work, a disability pension is awarded for an indefinite period of time. Otherwise, a rehabilitation benefit is awarded for a specific time period. The rehabilitation benefit is effectively a fixed-term disability pension. Despite its name, receipt of it is not conditional on participation in rehabilitation programs. There is no automatic retesting of disability status, except for new periods of rehabilitation benefits. More lenient eligibility criteria, including non-medical factors, are applied to applicants aged 60 years or more.

The entitlement period of regular UI benefits is about two years (100 weeks) for insured workers with sufficient work history. However, UI recipients who are old enough on the day when their regular benefits expire are entitled to extended benefits. In our observation period, the age threshold for the benefit extension was 57, 59 or 60, depending on the recipient's birth year. In addition, long-term unemployed workers born before 1950 were entitled to an unemployment pension at age 60. The unemployment pension scheme was abolished in 2005 for later birth cohorts. Extended UI benefits can be received until entitlement to unemployment pension (at age 60 for those born before 1950) or old-age pension (at age 63 for later cohorts) begins. The combination of regular and extended UI benefits plus possibly an unemployment pension is often dubbed as the "*unemployment tunnel*" (UT). The UT scheme provides a seamless flow of income until retirement for older unemployed individuals, and it has acted as a popular early retirement pathway.

2.2. Experience rating in disability insurance

A firm's DI premium rate is defined as a weighted sum of a flat-rate base premium and an experience-rated premium.¹⁰ The relative weight of the experience-rated premium depends on the firm's size: it increases linearly with the payroll (measured two years earlier) from 0% to 100% for firms with a payroll between 1.5 and 24 million in 2004 euros.¹¹ Thus, small firms with a payroll of 1.5 million euros or less are not subject to experience rating but only pay flat-rate premiums, whereas larger firms are at least partially responsible for the costs of disability pensions received by their former employees. It is worth emphasizing that the degree of experience rating (= the relative weight of the experience-rated part in the firm's DI premium) increases slowly with firm size. For example, the degree of experience rating is only 7% for a firm with a payroll of 3 million in 2004 euros (i.e. the payroll that is twice the size threshold of the large firm). So the smallest large firms are only marginally subject to experience rating.¹²

The experience-rated premium rate is determined by the estimated costs of new disability pensions awarded to the firm's employees two and three years ago. The cost of a disability pension equals the expected amount of disability pension benefits until the statutory old-age retirement age, so that younger disability pension recipients have larger

impacts on the experience-rated premium rate. When a firm's disability pension costs are equal to the average costs of firms with the same age structure, the experience-rated premium equals the base rate. At maximum, an experience-rated firm can earn a 90% discount on the base premium or be obligated to pay a 450% surcharge on top of the base premium. In 2007–2014, the base premium varied between 0.9% and 2% of the wage rate and the experience-rated premium between 0.09% and 11%.

In practice, large employers know their current DI premium rate but cannot estimate accurately how that would change if one additional worker were awarded a disability pension, because the future DI premiums depend on several unknown parameters. Using the same data used in this study, we estimated expected marginal costs of new benefit claims, i.e. discounted sums of expected increases in a firm's future DI premiums, in an earlier study (see Kyrrä and Tuomala, 2013). While these calculations were based on several assumptions, they give a rough picture of the magnitude of the DI costs due to experience rating. For example, for workers between the ages of 45 and 55 in firms with the degree of experience rating 50% or higher, we found that the average expected marginal costs can be close to the annual salaries of the considered workers.¹³ These calculations also show that the average expected marginal cost declines smoothly after the age of 55 as the statutory old-age retirement age approaches.

To the extent that experience rating in DI encourages firms to invest in disability-preventive measures, the experience-rating system should reduce inflows to vocational rehabilitation programs, sickness benefits, rehabilitation benefits, and disability pensions. However, conditional on the onset of a medical condition reducing work capacity, experience rating may increase inflows to other programs than disability pensions. This is because the experience-rated DI premium depends only on disability pension costs, not on rehabilitation benefit or rehabilitation allowance costs. This may induce large firms to encourage their workers with health problems to apply for a rehabilitation benefit or participate in vocational rehabilitation programs rather than to apply for a disability pension. If a worker's health status improves to the extent that he or she can return to work, their employer will avoid DI costs altogether. But even if the worker eventually ends up on a disability pension, their employer will gain from postponing that event, because the older the worker is at the time when the disability pension is awarded, the smaller the effect is on the firm's DI premium rate. Therefore, the overall effect of experience rating in DI on other health-related programs than disability pension is a priori ambiguous.

It should be stressed that large firms cannot easily escape DI costs by laying off workers whose health deteriorates. This is because a new disability pension claim affects the DI costs of the firms in which the claimant worked one and two calendar years prior to the year when the underlying medical condition was diagnosed. That is, if the claimant's medical condition was already diagnosed by a physician prior to the layoff, or if he or she receives the diagnosis in the year of the layoff or in the following two years, the former employer will be subject to the DI costs. If workers with health issues are more keen to apply for a disability pension when unemployed than when employed, experience rating in DI may even discourage layoffs. For the older unemployed who are eligible for the UT scheme, the incentive to apply for a disability pension can be weaker than for other unemployed

⁹ For both benefit types, full benefit is conditional on a loss of working capacity of at least 60% and partial benefit on a loss of 40% to 59%.

¹⁰ Here we only discuss experience rating in the private sector. Somewhat different experience-ratings systems are applied to municipal and governmental agencies.

¹¹ These payroll thresholds correspond approximately to firms with 50 and 800 full-time employees with the average wage.

¹² Also differences in the degree of experience rating around the upper payroll threshold of 24 million in 2004 euros are small. Therefore, it is not possible to apply the regression-discontinuity approach to estimate the causal effect of experience rating. Because the kinks in the degree of experience rating at the payroll thresholds are relatively small, also the regression-kink approach would be problematic due to the lack of statistical power.

¹³ In the earlier study, we did not find a significant effect of the expected marginal cost on the disability inflow, which may seem to be at odds with the results in the present study. However, at that time we only had data until 2010. As will be seen, the effect of experience rating in DI on the disability inflow is statistically significant only in the later years (e.g. after 2010 in Fig. 4, panel D). This is probably due to the gradual adoption of the experience-rating system and anticipatory behavior. In the earlier study, we also erroneously ignored the role of the UI costs. In the present study, we apply a much simpler and more transparent DID approach compared to the complex calculations of the expected marginal DI costs we performed in the earlier study.

because their income level is secured until old-age retirement due to the extension of the UI entitlement period. But their layoffs can become particularly costly for large employers due to coinsurance UI costs. Overall, it is not obvious how experience rating in DI should affect layoffs of workers with poor health.

2.3. Coinsurance in unemployment insurance

Regular unemployment benefits are financed by tax revenues and flat-rate insurance premiums, but large firms are partially liable for the costs of extended UI benefits paid to their former employees. When a benefit extension is awarded to a worker born in 1950 or later, the former employer, depending on its size, may have to pay a share of the estimated costs of the extended benefits as a lump sum payment to the Unemployment Insurance Fund. The cost of extended benefits is calculated assuming the worker will collect them until the statutory old-age retirement age irrespective of actual benefit duration.¹⁴ The share of the extended UI benefit costs the employer is directly liable for increases linearly from 0% to 80% as a function of the firm's payroll in the year preceding the layoff. The payroll thresholds for the minimum and maximum cost shares are the same as those that determine the degree of experience rating in DI, i.e. 1.5 and 24 million in 2004 euros. It follows that any two firms that differ in the degree of experience rating in DI also differ in the degree they are responsible for the extended UI benefit costs. Nevertheless, the effects of DI and UI costs can be separated because DI costs can arise from workers of all ages, but UI costs can arise only from the oldest workers born in 1950 or later.

The UI costs act like severance pay and should, therefore, discourage layoffs of older workers who are eligible for the UT scheme. Unlike the DI costs due to experience rating, the coinsurance UI costs are individual-specific, transparent and easily quantified in advance because they do not depend on any unknown parameters. Assuming the average wage and the maximum cost share of 80%, the employer's UI cost from laying off a worker at age 57 is roughly 20 times the worker's monthly wage. Thus, the UI costs can be relatively high compared to the expected DI costs. Therefore, from the viewpoint of economic incentives, it does not seem advisable to layoff older workers who are eligible for the UT scheme for health reasons.

3. The 2007 pension reform

Until 2006, DI and UI costs only applied to private-sector workers who were insured under the Employees' Pension Act (TEL). Over 90% of all workers in the private sector were covered by TEL. The only notable exception were blue-collar workers in construction, dock work, agriculture and forestry, who were insured under the Temporary Employee's Pensions Act (LEL) and accounted for some 6% of private-sector employment.¹⁵ Within firms, different worker groups may have

¹⁴ The employers were not liable for the costs of the extended benefits awarded to their workers born before 1950, but instead they were partially liable for the costs of unemployment pensions received by such workers. These birth cohorts qualified for an unemployment pension at age 60, had they been unemployed for at least two years. When that happened, the former employer, if large enough, had to make a lump sum payment (a given fraction of expected discounted unemployment pension costs) to the pension provided at the time when the unemployment pension was awarded. However, these unemployment pension costs did not apply to workers born before 1950 in the worker group analyzed in our analysis (i.e. a specific blue-collar worker group insured under a certain pension act).

¹⁵ The name of the pension act is a bit misleading: it was applied to all blue-collar worker groups in certain industries and certain occupation groups independently of whether they had a fixed-term or open-ended employment contract. Another exception were artists, journalists and persons working for households, who were insured under the Pension Act for Performing Artists and Certain Groups of Employees. That group accounted for less than 2% of private-sector employment. We do not consider that group in our analysis.

been insured under different pension acts. In construction firms, for example, white-collar workers were insured under TEL and blue-collar workers under LEL. Firms of all sizes only paid flat-rate DI and UI premiums for their LEL-insured workers. However, large firms were partially liable for the costs of disability and unemployment-related benefits received by their TEL-insured workers through experienced-rated DI premiums and coinsurance contributions.

In 2007, all private-sector pension acts were unified into a new Employees Pensions Act (TyEL).¹⁶ This reform did not affect the eligibility criteria or benefit levels of disability-related benefits, i.e. the content of DI from a worker's perspective, but it extended experience rating in DI to also cover blue-collar groups that were insured under LEL. As a consequence, large firms with a payroll exceeding 1.5 million in 2004 euros began to pay experienced-rated DI premiums also for their former LEL-insured workers, whereas smaller firms continued to pay flat-rate premiums as before. Due to a transitional provision, experience rating came into effect gradually over time: only disability pensions awarded to former LEL-insured workers whose medical condition was diagnosed in 2008 (partial effect) or later (full effect) have influenced the employer's DI premium rate. Another consequence of the 2007 reform was that large firms became responsible for the costs of extended UI benefits received by their former LEL-insured workers born in 1950 or later. These coinsurance UI costs applied to workers who were laid off in 2007 or later.

If small and large firms and their employees are sufficiently similar, we can exploit the 2007 reform to estimate the causal effects of experience rating in DI and coinsurance in UI using a difference-in-differences (DID) approach. Namely, we can use small firms and their employees as a comparison group, because the reform did not affect the DI and UI costs of small firms.

Fig. 2 illustrates eligibility for the UT scheme and the timing and coverage of employer costs in UI and DI in the former LEL industries. The shaded areas in the graph show the timing and coverage of UI and DI costs for large firms: experience rating in DI applies to all workers from 2008 onward (shaded area with dashed border), whereas coinsurance in UI only applies to workers born in 1950 or later at the ages when they are eligible for the UT scheme (shaded area with solid border). Since these areas only partially overlap, the effects of UI and DI costs are identified separately.¹⁷

4. Data

Our data is compiled by merging the registers of the Finnish Centre for Pensions and Statistics Finland. The Finnish Centre for Pensions is a semi-governmental body that co-ordinates the entire pension system and collects data from all pension providers for statistics and research purposes. Its databases include comprehensive records on job spells and earnings, as well as detailed information on sickness, rehabilitation, disability and pension benefits. The records also include the dates when a medical diagnosis was made that led to receipt of a rehabilitation benefit, rehabilitation allowance or disability pension (typically at the beginning of the sickness benefit period). This is important because the disability pension costs are assigned to firms where the benefit recipient worked one and two calendar years prior to the year of the medical diagnosis. We supplement these records by merging information on background characteristics from the databases of Statistics Finland. Complete data records are available for the years 1988–2015.

¹⁶ The government submitted the law change to parliament in April 2005 and it was accepted in May 2006.

¹⁷ Because the age threshold of the UT scheme varies between 55 and 58 depending on the birth year, we can distinguish the effect of UT eligibility from the age effect. In addition, we can identify the effect of replacing the unemployment pension with extra weeks of extended UI benefits by comparing workers born before and after 1950 at ages 60+. We are not really interested in this change but may want to control for its effect in the econometric analysis.

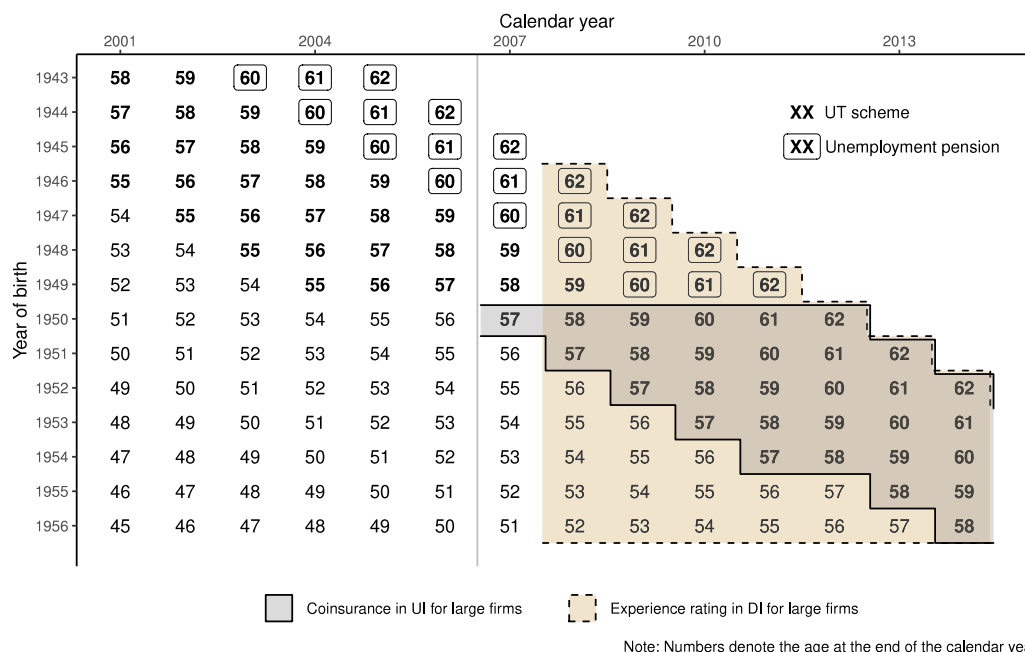


Fig. 2. Eligibility for the UT scheme and employers' UI and DI costs of their former LEL-insured workers by cohort, age and year. Note: Eligibility for the UT scheme (bold numbers) is defined as the lowest age at which a laid off worker is able to collect UI benefits, including regular benefits for the first two years and extended benefits thereafter, until entry into an unemployment pension (pre-1950 cohorts) or old-age pension (later cohorts). This age threshold varies between 55 and 58 depending on the birth year.

To be included in the analysis in year t , we require that a worker is 20–64 years old, he or she was employed and earned no less than 10,000 euros a year in the previous two years, and he or she was insured under LEL (before 2007) or would have been insured under LEL without the TyEL reform (2007 and later years).¹⁸ If the worker had more than one employer in these years, we require that over half of the wages were paid by a single firm in both years. These conditions make sure that a major part of the disability pension costs were assigned to one primary employer, if the worker received a diagnosis of a medical condition in year t that led to receipt of a disability pension.¹⁹ We consider firms that have at least five employees who satisfy the above conditions at least in one year in the pre- and post-reform periods. We exclude firms with fewer than five LEL-insured employees, because we are interested in employer behavior. We impose the time-period restriction in order to measure firm size prior to the reform. This is because some firms may have manipulated their size to avoid UI and DI costs, even though there is no evidence of such behavior (see the Supplementary Appendix). The sample restrictions drop a large number of small and unstable firms,²⁰ but our results are not very sensitive with respect to these restrictions as shown in the Supplementary Appendix.

¹⁸ Since our data contains the insurance type of each employment contract, we observe all LEL-insured workers until 2006. In the later years, we infer workers who would have been insured under LEL in the absence of the TyEL reform mainly on the basis of their industry, occupation and blue-collar status. In addition, we include all those workers who were insured under LEL and whose employment relationship continued with the same employer in the TyEL period. In the TyEL period, we exclude the employees of firms that did not have any LEL-insured workers before the reform.

¹⁹ To be specific, if a worker is awarded a disability pension in year $s \geq t$ based on the disability diagnosed in year t , the cost of this pension will be assigned to the firms in which he or she worked in years $t-1$ and $t-2$, when determining the firms' disability pension costs for year s . If there are more than one employer, the pension cost will be divided between the employers in proportion to the wages they paid to the worker in years $t-1$ and $t-2$.

²⁰ The restriction of at least 5 LEL-insured employees reduces the number of workers by 27% and the number of firms by 84%. In addition, the time-period restriction further reduces the number of workers and firms by 17% and 53%, respectively.

We analyze transitions from employment to sickness benefits, rehabilitation benefits, disability pension, vocational training and unemployment in 1999–2014. Since for the first weeks of sickness (typically one to three months) the applicant is fully compensated by the employer, we only observe relatively long spells of sickness benefits. When modeling rehabilitation benefit and disability pension inflows, we model the probability of the onset of a medical condition that leads to receipt of a rehabilitation benefit or disability pension, typically with a lag of one or two years. That is, we do not consider the year when the benefit was granted, but the year when the underlying medical condition was diagnosed because that determines which employers are liable for disability pension costs. When modeling layoffs and the inflow to sickness benefits, we further require that the worker was not unemployed and did not receive sickness benefits in the previous year, respectively.

It should be stressed that different outcomes are not mutually exclusive. For example, a worker may first collect a sickness benefit for one year, then a rehabilitation benefit for the next two years and finally transfer to a disability pension. If all these benefits were awarded for the same medical condition diagnosed at the beginning of sick leave, the worker became a recipient of sickness benefit, rehabilitation benefit and disability pension in the same year in our analysis. Likewise, a worker can become unemployed and disabled in the same year, if he or she received a medical diagnosis for a rehabilitation benefit or disability pension in the same year he or she was laid off.

5. Sample statistics

Table 1 reports sample statistics by period and firm size. A firm is classified as “large” if its average payroll in the pre-reform years exceeded 1.5 million in 2004 euros. Otherwise, the firm is classified as “small”. Our sample contains 95,442 individuals who worked in 2920 different firms. The vast majority of the firms (94%) are small, with payroll of 1.5 million or less in 2004 euros. Although only 6% of the firms are classified as large, these however employ almost half of all the workers. Roughly a quarter of the workers are employed in 21 firms whose payroll is at least 24 million in 2004 euros, and which are liable for the maximum share of UI and DI costs. The average degree of

Table 1
Summary statistics by firm size and time period.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Small firms			Large firms			DID	
	1999–2006	2007–2014	Change	1999–2006	2007–2014	Change	(6) - (3)	p-value
Panel A. Sample means across workers								
Age, years	40.8	42.2	1.48	42.6	43.7	1.09	−0.39	0.07
Female, %	5.1	5.3	0.22	4.3	4.5	0.23	0.01	0.99
Married, %	47.0	44.8	−2.25	51.2	47.9	−3.33	−1.08	0.13
Upper secondary education, %	58.9	64.7	5.84	59.8	69.2	9.39	3.55	<0.01
Tertiary education, %	5.1	5.4	0.30	4.3	6.1	1.79	1.49	<0.01
Kids under 7, %	20.3	18.4	−1.82	18.2	16.9	−1.32	0.50	0.30
Lives in owned flat, %	68.0	73.0	5.07	70.3	75.6	5.32	0.25	0.76
Lives in urban region, %	45.2	53.0	7.81	54.9	66.8	11.84	4.04	<0.01
Tenure, years	5.1	6.6	1.48	5.9	6.9	0.97	−0.51	0.01
Degree of experience rating, %	0.0	0.0	0.00	69.9	66.7	−3.18		
# observations	142,155	161,808		129,034	130,962			
# workers	40,784	42,780		29,965	29,817			
Panel B. Sample means across firms								
Construction, %	66.0	68.0	2.00	48.1	47.9	−0.17	−2.17	0.23
Cargo handling, %	0.7	0.8	0.10	7.8	9.1	1.31	1.21	0.05
Agriculture and forestry, %	13.9	14.9	1.00	7.7	7.6	−0.09	−1.09	0.33
Other industry, %	19.5	16.4	−3.10	36.5	35.5	−1.05	2.05	0.30
Degree of experience rating, %	0.0	0.0	0.00	24.0	24.2	0.14		
# observations	14,235	14,948			1,200	1,148		
# firms	2,751	2,751		169	169			
Panel C. Average annual inflow rates across workers, %								
Sickness benefits	5.00	5.05	0.05	6.71	5.92	−0.79	−0.84	<0.01
Vocational rehabilitation	0.14	0.25	0.10	0.15	0.41	0.26	0.16	<0.01
Rehabilitation benefits	0.52	0.51	−0.01	0.56	0.54	−0.02	−0.01	0.82
Disability pensions	0.81	0.67	−0.14	1.06	0.74	−0.32	−0.19	<0.01
Layoffs	3.23	3.25	0.02	3.68	3.81	0.12	0.10	0.72
– UT-eligible workers	5.22	4.79	−0.43	6.39	5.43	−0.96	−0.53	0.49
– Non-UT-eligible workers	3.04	3.06	0.02	3.32	3.55	0.24	0.22	0.47

Notes: A firm is classified as large if its average payroll was more than 1.5 million in 2004 euros in the pre-reform years. Tenure is top-coded at 11 years because completed employment records are available from 1988 onward. Degree of experience rating refers to the relative weight of the experience-rated premium rate in the firm's DI premium rate formula, and it is computed using the average payroll in the pre-reform years. It increases linearly from 0% to 100% from a payroll of 1.5 million to a payroll of 24 million euros, and it is 0% for smaller firms and 100% for larger firms. Column 7 reports difference-in-differences estimates for worker and firm characteristics, i.e. the difference between columns 6 and 3. The associated *p*-value in column 8 is obtained by regressing an outcome variable (at worker or firm level) on the time period, large-firm dummy and their interaction, using a variance-covariance matrix robust to individual- and firm-level clustering (Panels A and C) or firm-level clustering (Panel B).

experience rating in DI — the weight of the experience-rated premium in the DI formula — is 24% among the large firms but as much as 70% among their employees.

Over half of the firms operate in the construction sector. Most sample workers are men with upper secondary or lower education. The workers of the small and large firms are relatively similar in terms of their background characteristics, and the sample composition changes similarly over time, as seen in columns 3 and 6 in panel A. In the last two columns, we test changes in the sample composition formally: column 7 shows the difference between columns 6 and 3, and column 4 reports the *p*-value of the test that this difference does not differ from 0. The age, gender and family structure of the workforce as well as the industry composition have changed similarly over time in firms of different sizes, but there are also some statistically significant changes in the background characteristics. The shares of workers with upper secondary and tertiary degrees have been increasing over time in the small and large firms, but the pace has been faster in the large firms. In addition, the average tenure increased by half a year less while the share of workers living in urban regions increased by four percentage points more in large firms compared to small firms between the two periods.

Panel C shows average annual exit rates from employment to different types of benefits. Many workers ended up in sick leave or became unemployed at some point in the observation period. This is not surprising because, unlike receipt of disability-related benefits, these events are relatively common in all age groups and may be experienced several times during the observation period. The layoff rate of older workers who are eligible for the UT scheme is clearly higher than that

of younger workers in both small and large firms, but the difference is larger in the large firms. While the overall layoff rate and the layoff rate of younger workers are stable or only increase marginally over time, the layoff rate of UT-eligible workers declines by 0.43 percentage points in small firms and by 0.96 percentage points in large firms, but the difference in these changes is not statistically significant.

The rehabilitation benefit inflow rate remains relatively stable over time in both groups, but the development of other health-related inflow rates differs significantly between small and large firms. The sickness benefit inflow decreases in the large firms but remains roughly constant in the small firms. While the inflow rate to vocational rehabilitation programs is very small, it almost triples in the large firms and roughly doubles in the small firms. In 2004, vocational rehabilitation was reformed. The focus of rehabilitation measures shifted from disability benefit recipients to those who are still in the workforce, and occupational rehabilitation became a subjective legal right for workers who are still in employment but at risk of work-capacity loss. The objective of the 2004 reform was to advance the start of occupational rehabilitation and to improve the effectiveness of rehabilitation measures. Because the changes applied to all workers irrespective of their employer, this reform probably explain why the inflow to vocational rehabilitation programs increases in firms of all sizes.

In the large firms, the disability pension inflow rate decreases by 0.32 percentage points between the two periods, compared to a significantly smaller decline of 0.14 percentage points for the small firms. Overall, the numbers reported in columns 7 and 8 suggest that the 2007 reform reduced the sickness benefit inflow by 0.84 percentage points and the disability pension inflow by 0.19 percentage points and

increased participation in vocational rehabilitation programs by 0.16 percentage points in the large firms. However, these estimates should be interpreted with caution, as they may be subject to some bias. In the next section, we find evidence of excess layoffs and sharp drops in the health-related benefit inflows among large firms in 2006, suggesting that the DID estimates in column 7 are affected by anticipatory behavior of large employers one year before the reform came into effect. In the subsequent analysis, we examine the time pattern of the reform effects, explicitly allow for anticipatory effects, assess the validity of the underlying parallel trend assumption, and estimate distinct effects for experience rating in DI and coinsurance in UI.

It should be stressed that the worker group studied is characterized by relatively high disability and layoff risks. In the Supplementary Appendix, we compare the disability and unemployment outcomes of TEL- and LEL-insured workers in 1999–2006. The LEL-insured workers have about a 30% higher risk of becoming disability benefit recipients and a 70% higher layoff probability than the TEL-insured workers. Among the LEL-insured workers, the most common causes of disability are musculoskeletal and connective tissue diseases (45%), while mental health and behavioral disorders are relatively rare (9% compared to 21% among the TEL-insured workers).

6. Results

6.1. The effect of the 2007 reform

Fig. 3, panels A to E depict the average inflow rates to different benefits by year and firm size. Panel F shows the share of employees who remain employed with the same employer they worked for in the past two years. To assess the statistical significance of the changes between the firm-size groups, we estimate linear probability models of the following form:

$$Y_{ijt} = \alpha + \delta \text{Large}_j + \sum_{s=1999}^{2014} \varphi_s \mathbf{1}\{t=s\} + \sum_{s=1999}^{2014} \gamma_s (\text{Large}_j \cdot \mathbf{1}\{t=s\}) + \varepsilon_{ijt}, \quad (1)$$

where i indexes worker, j indexes firm, and t indexes time. Y_{ijt} equals 1 if the worker experiences the event of interest in year t , and 0 otherwise. Large_j is a dummy variable for the employees of the large firms. $\mathbf{1}\{\cdot\}$ is the indicator function, and φ_s are year fixed effects. We choose 2005 as the reference period and impose the restriction $\varphi_{2005} = \gamma_{2005} = 0$. The parameters of interest are γ_s , which capture differences in the outcomes between the large and small firms in year s , net of the difference in 2005. The estimates of these DID effects along with their cluster-robust 95% confidence intervals are shown in Fig. 4.

In Fig. 3, the curves for the small and large firms move in parallel until 2005 (until 2003 in the case of layoff risk).²¹ Except for the layoff rate in 2003, the differences in the inflow rates between the small and large firms remain roughly constant over the years 1999–2005, as the 95% confidence intervals contain 0 each year in Fig. 4. Thus, the parallel trend assumption of the DID approach seems to be valid.

In the large firms, the inflow rates to sickness benefits and disability pensions (Fig. 3, panels A and D) drop, whereas the inflow rate to vocational rehabilitation programs (panel B) increases in the post-reform period compared to small firms. The inflow to vocational rehabilitation programs increases over time also in small firms due to the vocational rehabilitation reform in 2004. For the sickness benefit

inflow, all the differences in the post-reform years are statistically significant at the 5% level, as seen in Fig. 4, panel A. For the inflows to disability pensions and vocational rehabilitation programs, only a few of the annual differences are statistically significant in the same period (Fig. 4, panels B and D) due to the much smaller numbers of transitions. However, the average effects over the post-reform years are significant, as seen in Table 1, panel C. It is difficult to see any systematic trend in the differences in the layoff risk or rehabilitation benefit inflow rate between small and large firms (Fig. 4, panels C and E). Likewise, the probability of remaining with the same employer changes rather similarly over time except for a disproportionately large drop in small firms during the financial crisis in 2007 and 2008.

Unlike in small firms, in large firms the inflows to rehabilitation and sickness benefits and to disability pensions declined sharply in 2006 (Fig. 3, panels A, C and D). In the same year, the layoff rate in the large firms was rather high compared to the small firms (Fig. 3, panel E), even though the difference is not significantly larger than in 2005 (Fig. 4, panel E). A closer look at the inflow rates by age groups reveals that in the large firms the layoff rate of older workers eligible for the UT scheme increased from 6.2% in 2005 to 9.0% in 2006. At the same time, the layoff rate of younger workers in large firms and the layoff rates of both worker groups in small firms declined. These observations suggest that some large firms exploited the UT scheme to get rid of their older workers with a high disability risk by laying them off one year before the reform came into effect. In doing so, the employer avoided DI costs that would have been realized if some of their UT-eligible workers had been awarded a disability pension after 2007, with or without being laid off first.

6.2. Separating the effects of UI and DI costs

The DID estimates in Table 1, panel C and in Fig. 4 describe the joint effect of employers' DI and UI costs. These costs may have opposite effects on some outcomes, potentially overriding each other within the UT-eligible worker group. To separate the two distinct effects, we estimate the following linear probability model:

$$Y_{ijt} = \mu + \sum_{s=2000}^{2014} \varphi_s \mathbf{1}\{t=s\} + \alpha_1 UT_{it} + \alpha_2 \text{Large}_j + \alpha_3 \text{Born}_i^{1950+} + \delta_1 (UT_{it} \cdot \text{Born}_i^{1950+}) + \delta_2 (\text{Large}_j \cdot UT_{it}) + \delta_3 (\text{Large}_j \cdot \text{Born}_i^{1950+}) + \lambda (\text{Large}_j \cdot UT_{it} \cdot \text{Born}_i^{1950+}) + \sum_{s=2006}^{2014} \gamma_s (\text{Large}_j \cdot \mathbf{1}\{t=s\}) + X_{ijt}\beta + \varepsilon_{ijt}. \quad (2)$$

where UT_{it} is a dummy variable for workers who are eligible for the UT scheme in year t , Born_i^{1950+} is a dummy variable for those born in 1950 or later, and X_{ijt} is a vector of controls for age, gender, education level, marital status, tenure, living region, having a young child, home ownership and industry.²²

Time variation in the outcomes across the employees of small firms identifies the year fixed effects φ_s . As illustrated in Fig. 2, the age threshold of the UT scheme is 55 for individuals born before 1950, 57 for those born in 1951–54, and 58 for later cohorts. This variation across birth cohorts identifies the effect of UT eligibility, α_1 , in the presence of age and year effects. The interaction term $\text{Large}_j \cdot UT_{it}$ allows the effect of UT eligibility to be different in large and small firms. The interaction term $UT_{it} \cdot \text{Born}_i^{1950+}$ equals 1 for those who qualify for extended UI benefits after the entitlement period of regular

²¹ Note that the declines in the disability pension and rehabilitation benefit inflow rates in the last years of the observation period are in part spurious, because we can observe these outcomes (i.e. the timing of the underlying medical diagnoses) only when benefits were awarded in 2015 or earlier. Because this problem applies to the employees of both small and large firms, it should not distort our comparisons.

²² For age effects we use dummy variables for each age between 51 and 60, and for those less than 40 years, 41–50 years, and over 60 years. We model the tenure effect using the second order polynomial. For the other control variables we use dummy variables for the same categories as used in Table 1.

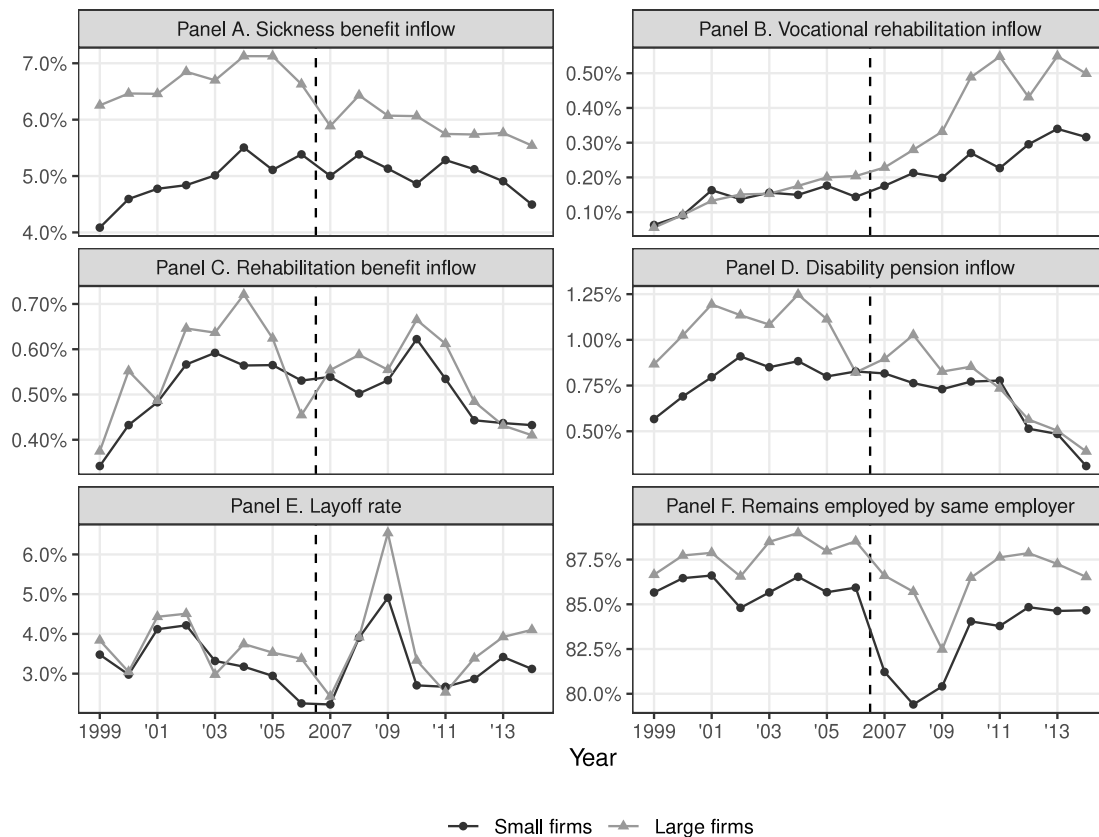


Fig. 3. Inflow rates by year and firm size.

UI benefits but not for the unemployment pension, and 0 for others (see Fig. 2). Thus, δ_1 is the effect of replacing the unemployment pension with extra weeks of UI benefits from age 60 upwards, and it is identified by comparing the UT-eligible employees of small firms who were born before 1950 and those born in 1950 or later at the same age.

The parameters of primary interest are λ , the effect of employer's UI costs through the lump sum coinsurance payments, and γ_s , the effect of employer's DI costs through the experience-rating system. The effect of DI costs is allowed to vary over time, starting from 2006, i.e. one year before the reform came into effect, in order to allow for anticipatory behavior and to capture the gradual implementation of experience rating. The findings in the previous section highlight the importance of allowing for anticipatory effects. Treating 2006 as part of the pre-reform period would lead to biased estimates due to anticipatory behavior of large employers. To increase statistical power, we assume that the effect of DI costs is constant in the years when the experience rating system was fully implemented and, therefore, we impose the restriction $\gamma_{2009} = \gamma_{2010} = \dots = \gamma_{2014}$. As we did not find significant differences between large and small firms in 1999–2005 except for a difference in the layoff rate in 2003 (see Fig. 4), we assume that the reform had no effect prior to 2006. This assumption is formally tested in the Supplementary Appendix.

The effects of DI and UI costs are identified from differences in the outcomes over time between workers employed by large and small firms. The two effects should be separately identified, because the UI costs can only arise from workers born in 1950 or later who enter unemployment after reaching the age threshold of the UT scheme, while the DI costs can arise from all disability pension recipients irrespective of their birth year and age (see Fig. 2). We assume that the introduction of experience rating and coinsurance costs for large firms did not affect the outcomes of the employees of small firms. There is no reason to suspect the validity of this assumption. However, we also

need to assume that the employer's coinsurance UI costs for UT-eligible workers born in 1950 or later have no displacement effect on the layoff rates of other workers within the same firm. It is possible that some large employers shifted layoffs from older workers who became subject to coinsurance UI costs towards younger groups after the reform. If so, the assumption will not hold and γ_s will not only capture the spillover effect of experience rating in DI on layoffs but also the displacement effect of UI costs within large firms. Since the latter effect is expected to be positive, the estimates of γ_s for the layoff risk can be biased upward. However, the potential bias is likely to be small, provided that only 9% of the employees of large firms in the post-reform period are subject to coinsurance UI costs. In Table 1, panel C, the layoff rate of younger workers who are not eligible for the UT scheme increased 0.22 percentage points more in large firms than in small firms between the pre- and post-reform periods. While this difference is small compared to the average layoff risk and statistically insignificant (p -value 0.47), it is consistent with a small displacement effect, which should be kept in mind when interpreting the results for the layoff risk.

The estimated effects of the key variables, in percentage points, are reported in Table 2. From column 1 we see that the likelihood of starting a sickness benefit period is 1.6 percentage points higher for individuals who are employed in large firms than for those working in small firms. The inflow rates to other benefits do not differ notably between large and small firms. As seen in column 5, eligibility for the UT scheme increases the layoff risk by 0.8 percentage points in small firms and by 2.3 percentage points ($= 0.843 + 1.438$) in large firms. It seems that firms often target layoffs at older workers whose income level is secured up to old-age retirement, and this practice is particularly common among large firms. Replacing the unemployment pension with extra weeks of UI benefits has no effect on the layoff risk, as the coefficient on the interaction term between UT eligibility and those born in 1950 or later does not differ from 0 at the conventional risk levels.

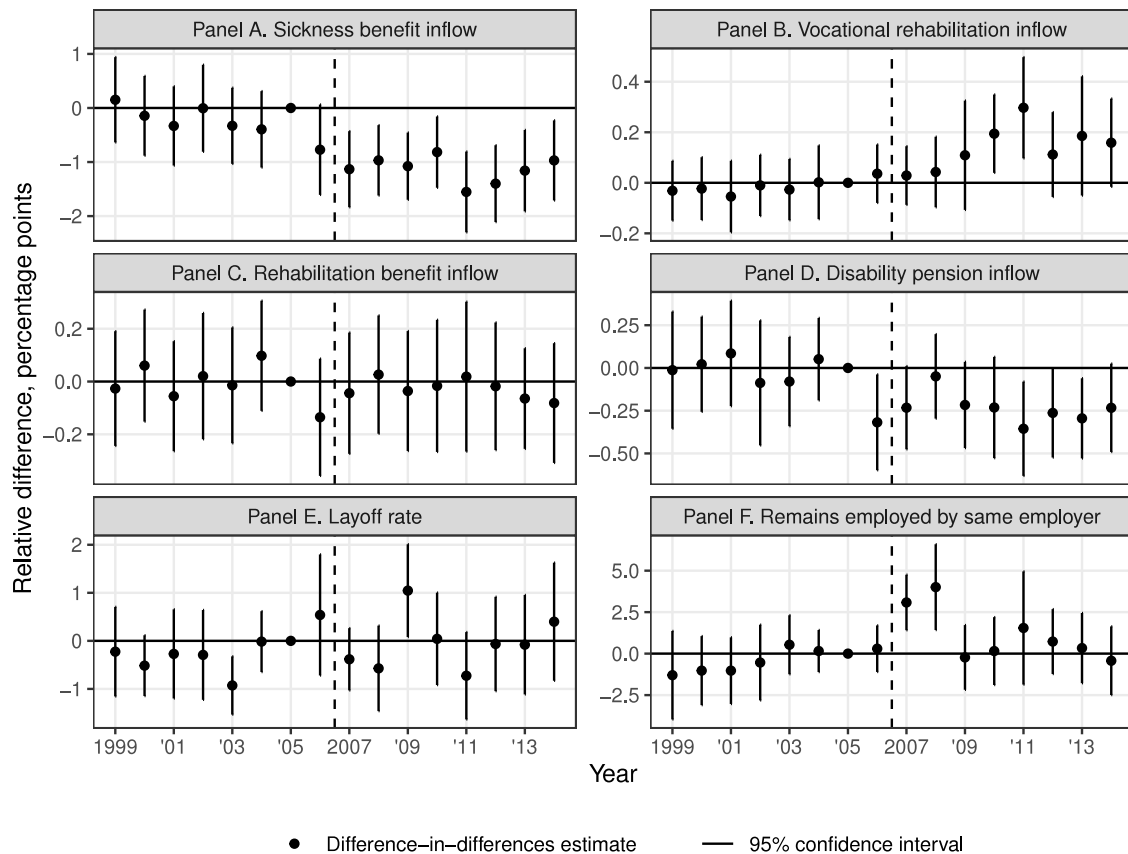


Fig. 4. Difference-in-differences estimates of the effect of the 2007 reform.

Coinurance UI costs reduce the layoff rate of older workers by 1.4 percentage points (column 5). This effect offsets the difference of 1.4 percentage points in the effect of UT eligibility between large and small firms, so that the layoff risk does not differ between UT-eligible workers in small and large firms when the large firms are subject to coinurance UI costs. It is noteworthy that UI costs do not increase the take-up of other benefits, given that the coefficient on the triple interaction term is negative, yet insignificant, for all health-related benefits. These results imply that coinurance in UI eliminates excess layoffs of older workers in large firms that would otherwise occur due to the UT scheme, without having adverse spillover effects on inflows to other types of benefits.

The introduction of experience rating in 2007 started to reduce sickness benefit claims from the first year onward: in large firms, the inflow to sickness benefits dropped by 0.7–1.0 percentage points in 2007–2014 compared to small firms (column 1). The inflow declined by 0.6 percentage points already in 2006. Although the effect in 2006 is imprecisely estimated and not statistically significant, it suggests that the estimated effect over the first post-reform years may also reflect some anticipatory behavior. But the strong effect over the later years implies that experience rating induces employers to invest in preventive measures that reduce the onset of health problems at the workplace and, therefore, the inflow to sickness benefits. The average duration of sickness benefits increased by over 20% in large firms compared to small firms after the reform (see the Supplementary Appendix). That is, large employers have mainly succeeded in eliminating short sickness spells, which explains much of the strong effect on the sickness benefit inflow in the years 2007–2014.

In 2009–2014, the inflow to vocational rehabilitation programs increased by 0.2 percentage points more in large firms than in small firms (column 2), which is a large relative effect given that the average annual inflow is only 0.2%. This finding suggests that experience

rating induces employers to direct workers who have difficulties in performing their current job tasks due to health problems to vocational rehabilitation programs. Large employers should have a strong financial incentive to do so, because the vocational rehabilitation programs are financed by the pension institutions and because the programs can reduce disability pension claims by enabling workers with disabilities to switch to more suitable jobs that allow them to continue working.

We find no evidence of experience rating in DI having significant effects on the inflow to rehabilitation benefits (column 3) or on layoffs (column 5). The point estimate of the layoff effect in 2006 is large, pointing to an increase of 0.9 percentage points in the layoff rate among employees of large firms, but the effect is imprecisely estimated and, therefore, statistically insignificant at the conventional risk levels. In 2006 and 2007, the inflow to disability pensions in large firms declined by 0.3 and 0.2 percentage points, respectively, compared to small firms (column 4). As discussed in the previous section, some large firms may have laid off their older workers with high disability risks just before the reform came into effect. This kind of anticipatory behavior could lead to a spurious negative effect on the disability pension inflow (and other health-related outcomes) in the short run and may explain the effects in 2006 and 2007. However, we also find a drop of 0.2 percentage points in the disability pension inflow over the years 2009–2014 in large firms compared to small firms. Since this estimate is less likely to be affected by anticipatory behavior, it seems evident that experience rating reduces disability pension claims. There is also a positive effect on the probability of remaining employed with the same firm (column 6).

It is worth emphasizing that the effects of experience rating on health-related outcomes come without adverse spillover effects on the layoff risk. This conclusion would not change even if the effect of experience rating on the layoff risk were subject to some bias due to a displacement effect of UI costs within large firms. This is because the displacement effect is expected to be positive.

Table 2
Effects on employment exits.

	(1) Sick leave	(2) Vocational rehab	(3) Rehab benefits	(4) Disability pension	(5) Layoff	(6) Same employer
Mean outcome, %	5.614	0.236	0.534	0.811	3.471	85.46
Large	1.616*** (0.525)	0.035 (0.042)	−0.052 (0.169)	0.084 (0.252)	−0.048 (0.449)	−0.412 (1.129)
UT eligibility	−0.013 (0.557)	0.053 (0.047)	0.030 (0.187)	0.615** (0.269)	0.843* (0.432)	−3.360*** (0.680)
Born 1950+	0.354 (0.424)	0.099*** (0.036)	0.152 (0.143)	−0.067 (0.190)	−0.079 (0.334)	−1.217** (0.538)
Large x UT eligibility	0.170 (0.568)	−0.046 (0.040)	0.019 (0.188)	0.323 (0.290)	1.438** (0.730)	−1.547 (1.093)
Large x Born 1950+	−0.256 (0.514)	−0.006 (0.041)	0.079 (0.170)	0.015 (0.256)	0.342 (0.390)	1.024 (0.721)
UT eligibility x Born 1950+	0.630 (0.559)	−0.126*** (0.049)	0.095 (0.172)	−0.322 (0.281)	0.623 (0.448)	2.424*** (0.740)
Effect of DI costs						
Large x Year 2006	−0.554 (0.468)	0.053 (0.047)	−0.146* (0.080)	−0.326*** (0.096)	0.938 (0.600)	0.483 (0.796)
Large x Year 2007	−0.898*** (0.327)	0.046 (0.045)	−0.049 (0.082)	−0.209** (0.087)	0.012 (0.344)	3.497*** (0.805)
Large x Year 2008	−0.694** (0.323)	0.060 (0.051)	0.026 (0.077)	0.013 (0.091)	−0.209 (0.422)	4.506*** (1.277)
Large x Year 2009–2014	−0.981*** (0.256)	0.194*** (0.063)	−0.029 (0.055)	−0.173*** (0.048)	0.132 (0.403)	2.084** (0.929)
Effect of UI costs						
Large x UT eligibility x Born 1950+	0.061 (0.664)	−0.029 (0.061)	−0.167 (0.202)	−0.257 (0.343)	−1.394** (0.694)	0.747 (1.122)
Number of observations	533,408	563,959	563,959	563,959	532,376	563,959

Notes: All models include year dummies and controls for age, gender, education, marital status, tenure, region of residence, having a child under age 7, home ownership and industry. Standard errors clustered at individual and firm levels in parenthesis.

*Significance at the 10% level.

**Significance at the 5% level.

***Significance at the 1% level.

Given that the expected annual disability pension inflow in large firms over the post-reform years without the reform is 0.92%, the results imply that experience rating reduced the disability pension inflow by 19% among the employees of large firms and by 9% among all employees, provided that the large firms employed 46% of former LEL employees. The former effect is close to some estimates for the Netherlands. Namely, [van Sonsbeek and Gradus \(2013\)](#) estimate that the introduction of experience rating in the late 1990s reduced the disability inflow by 13% in the Netherlands, while [Koning \(2009\)](#) finds that the disability inflow decreased by 15% in the Dutch firms that experienced an unanticipated premium increase compared to the firms with no change in their premium rates. But [de Groot and Koning \(2016\)](#) find a much smaller effect when they studied the removal of experience rating from small Dutch firms.

In summary, the results imply that the introduction of experience rating in DI reduced the inflow to sickness and disability pension benefits, and increased participation in vocational rehabilitation programs. In addition, the adoption of coinsurance in UI reduced the layoff rate of the oldest workers with a particularly high risk of unemployment due to their eligibility for the UT scheme. There is no evidence of adverse spillover effects: experience rating in DI does not increase layoffs, nor does coinsurance in UI increase the inflow to health-related programs.

6.3. Selective hiring

The aim of experience rating in DI is to encourage employers to invest in preventive measures that reduce the onset of health problems, as well as to accommodate workers with a medical condition that reduces their working capacity, so that they can continue working. However, experience rating may also discourage employers from hiring workers with high disability risks. That way, contrary to its primary goal, the experience rating system may worsen the employment prospects of job seekers with poor health. To minimize the risk of future disability

pension claims, hiring firms may require physical examinations of job applicants or favor younger job applicants over older ones. Selective hiring can reduce health problems in large firms subject to experience rating and, therefore, explain part of the effects of experience rating in DI in [Table 2](#). To assess this possibility we can compare new workers in small and large firms before and after the reform.

In [Table 3](#) we report the sample means of background characteristics for recently hired employees. Unfortunately, our data does not include good measures of health. The best measure is the time spent on sickness benefits before recruitment. Average days on sickness benefits among recently hired workers remain stable over time in large firms but increase slightly in small firms. The difference between these changes in the large and small firms is small, −0.9 days, but statistically significant at the 5% level (columns 7 and 8), suggesting that large employers may have been more cautious in hiring after the reform. The average age of recently hired workers increases by 0.2 years in large firms, which does not differ significantly from the drop of 0.8 years in small firms. Likewise, the share of new workers aged 50 and older increases more in large firms than in small firms, but the difference is not statistically significant. Also in most other cases, the composition of recently hired employees changes in the same direction by roughly the same amount in the large and small firms. One exception is the share of those with tertiary education, which increases in large firms but decreases in small firms. This group is very small, accounting for less than 7% of new hires as well as of all employees in both firm-size groups, so it cannot have a notable impact on the average outcomes. In the light of these differences, our estimates of the inflow effects are unlikely to be significantly affected by selective hiring after the reform, although the small difference in the sickness benefit days between large and small firms is a little worrisome.

As an another check, we re-estimate linear probability models using data only on workers who have been continuously working for the same employer for at least eight years. In that sample, all workers in all years

Table 3
Sample means for recently hired employees by firm size and period.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Small firms			Large firms			DID	
	1999–2006	2007–2014	Change	1999–2006	2007–2014	Change	(6) - (3)	p-value
Age, years	38.0	37.2	−0.81	38.8	39.0	0.19	1.00	0.38
Age ≥ 50, %	18.2	20.5	2.30	20.6	24.6	4.05	1.75	0.51
Female, %	4.9	5.3	0.47	4.1	4.1	0.00	−0.47	0.46
Married, %	38.1	33.7	−4.38	41.5	37.7	−3.85	0.53	0.82
Upper secondary education, %	60.7	68.0	7.35	62.8	71.6	8.78	1.42	0.22
Tertiary education, %	5.6	4.3	−1.33	5.4	6.1	0.73	2.06	<0.01
Kids under 7, %	19.3	19.1	−0.25	19.0	18.6	−0.48	−0.23	0.80
Lives in owned flat, %	57.5	61.0	3.53	60.1	67.7	7.63	4.11	0.09
Lives in urban region, %	49.3	60.1	10.85	53.7	67.5	13.75	2.91	0.06
Sickness days in year $t - 1$	4.5	5.3	0.86	4.8	4.7	−0.03	−0.89	0.04
# observations	23,918	8,037		17,766	8,711			
# workers	22,633	7,837		16,292	8,235			

Notes: The figures are for recently hired individuals who remained in the firm for at least for two years. A firm is classified as large if its average payroll was more than 1.5 million in 2004 euros in the pre-reform years. Column 7 reports the difference-in-difference estimates for worker characteristics, i.e. the difference between columns 6 and 3. The associated p-value in column 8 is obtained by regressing an outcome variable on the time period, the large-firm dummy and their interaction, using the variance-covariance matrix robust to firm-level clustering.

Table 4
Effects on employment exits for employees with at least 8 years of tenure.

	(1)	(2)	(3)	(4)	(5)	(6)
	Sick leave	Vocational rehab	Rehab benefits	Disability pension	Layoff	Same employer
Mean outcome	5.501	0.243	0.57	1.000	2.176	90.413
Effect of DI liability						
Large x Year 2006	−1.333** (0.570)	0.064 (0.065)	−0.167 (0.114)	−0.525*** (0.151)	1.058 (0.920)	−0.790 (0.819)
Large x Year 2007	−1.405*** (0.524)	0.044 (0.061)	−0.189 (0.119)	−0.417*** (0.152)	−0.023 (0.404)	1.817 (1.138)
Large x Year 2008	−0.778* (0.442)	−0.022 (0.067)	0.005 (0.116)	0.033 (0.159)	0.049 (0.631)	2.966** (1.324)
Large x Year 2009–2014	−1.304*** (0.342)	0.236*** (0.080)	−0.079 (0.084)	−0.219** (0.092)	0.380 (0.436)	1.313** (0.581)
Effect of UI liability						
Large x UT eligibility x Born 1950+	0.169 (0.952)	−0.120 (0.102)	0.027 (0.283)	0.340 (0.492)	−0.768 (0.770)	−0.796 (1.426)
Number of observations	235,514	249,826	249,826	249,826	241,641	249,826

Notes: All models include year dummies and controls for age, gender, education, marital status, tenure, region of residence, having a child under age 7, home ownership and industry. The standard errors clustered at individual and firm levels in parenthesis.

*Significance at the 10% level.

**Significance at the 5% level.

***Significance at the 1% level.

were hired before the 2007 reform, so that possible changes in hiring practices due to the reform cannot affect the results. The results for the effects of DI and UI costs from this subsample are shown in Table 4. For all outcomes the effects of DI costs are similar to those reported in Table 2, implying that the effects of experience rating in DI are not driven by selective hiring. If anything, the effects of experience rating are somewhat stronger for long-tenured workers. In contrast, the effect of UI costs on the layoff risk is only half of the corresponding baseline estimate in Table 2 (−0.8 vs. −1.4 percentage points), and statistically insignificant. However, since the average layoff risk is much smaller among long-tenured workers than among all workers (2.2% vs. 3.5%), the relative effect is in line with our baseline estimate (a 35% vs. 40% decline in the layoff rate due to coinsurance in UI). Obviously the comparisons of the UI cost effects are somewhat vague and inconclusive because the effects are imprecisely estimated in Tables 2 and 4.

6.4. Robustness and heterogeneity

We have checked the robustness of our results for the effects of experience rating in DI and coinsurance in UI in various ways. We discuss these findings briefly here, but the detailed results are reported in the Supplementary Appendix.

We formally test the parallel trend assumption by adding interactions between firm-size and year effects over the pre-reform years to the model outlined in Eq. (2). Since only one out of 36 coefficients on these interaction terms across models for different outcomes differs from 0 at the conventional risk levels, the parallel trend assumption seems to be valid. In addition, using data from the pre-reform period, we consider a placebo reform in 2003 and find no significant placebo effects of DI and UI costs except the effect of UI costs on the inflow to vocational rehabilitation programs. However, this only significant placebo effect can be attributed to a single outlier observation.

All the results in Section 6.2 are based on the models that include controls for background characteristics. Our results are not sensitive with respect to the exclusion of these control variables (apart from age) or to the inclusion of firm fixed effects. In the main analysis, we include firms with at least five LEL-insured workers. The results do not change notably when we consider firms with at least three LEL-insured employees. When only firms with at least ten LEL-insured employees are considered, the results remain otherwise similar but the effect of UI costs on the layoff risk becomes somewhat smaller in absolute terms (−1.031 vs. −1.394) and statistically insignificant.

In Section 5, we found some differential changes in the composition of the workforce of large and small firms over time. In particular,

the average education level has been increasing slightly faster in large firms than in small firms. This may affect the results as education is potentially correlated with health risks. Therefore, we estimate models separately for workers with upper secondary or higher education and for those with lower education. The results are rather similar for both education groups, implying that our main results are not driven by differential education trends. In addition, we estimate models separately for construction workers and for other workers. The results are otherwise similar but experience rating in DI increases the inflow to vocational rehabilitation programs only among construction workers. When only workers aged 45 and over are included in the analysis, the results do not change notably except that the effect of UI costs on the layoff risk becomes smaller and insignificant (-0.956 vs. -1.394).

In the main analysis, we treat large firms as a single group, ignoring the fact that the degree of experience rating and coinsurance varies across these firms. Firms whose payroll is less than 24 million but more than 1.5 million in 2004 euros are only partially subject to the DI and UI costs of their employees. If we exclude these firms and only include the 21 largest firms that are fully covered by experience rating in DI and coinsurance in UI in the group of large firms, we will find somewhat stronger effects for both types of costs than those reported in Table 2. Analogously, when we only consider smaller large firms that are only partially covered by experience rating and coinsurance, we find smaller and for some outcomes insignificant effects of DI and UI costs. These results imply that lower degrees of experience rating and coinsurance are associated with weaker responses by employers, as expected.

In summary, our results for the effects of DI costs are highly robust and relatively similar across different subsamples and specifications. However, the effect of UI costs on the layoff risk is less robust and generally rather imprecisely estimated due to the relatively small group of workers in the large firms who became subject to coinsurance in UI after the reform. In some subsamples, the estimated effect is clearly smaller in absolute terms than our baseline estimate of -1.394 and does not differ significantly from 0. But also in these cases, the estimated effect is around -1.0 and hence in the same ballpark with our baseline estimate.

7. Concluding remarks

The aim of experience rating in DI is to promote preventive health and safety practices at the workplaces, and to encourage employers to accommodate their employees with health-related work limitations. This study provides evidence of such effects. We find declines in the sickness benefit and disability pension inflow rates in large firms when they became exposed to experience rating. Our estimate of the 19% decline in the disability pension inflow among experience-rated firms is close to the estimated effects of experience rating in the Netherlands in Koning (2009) and van Sonsbeek and Gradus (2013). We also find that the employees of the large firms are more likely to participate in vocational rehabilitation programs in the post-reform years. We show that none of these effects is driven by selective hiring. We find no evidence of negative spillover effects on layoffs and, therefore, experience rating also increases the probability of continuing working for the same employer.

Our findings differ from the previous results of Kyyrä and Paukeri (2018), who did not find significant effects for other worker groups in Finland. In this study, we analyze a rather special group of blue-collar workers in construction, dock work, agriculture and forestry that is characterized by a higher-than-average disability risk. One possibility is that employers have better possibilities to influence disability outcomes when the initial level of disability inflow is high. This hypothesis is consistent with the findings of van Sonsbeek and Gradus (2013), who found larger effects of experience rating in sectors with higher disability risks (e.g. the construction sector) in the Netherlands. Employers' possibilities to prevent transitions to disability benefits probably also depend on what types of health risks their workers are subject to.

For the worker group analyzed in this study, musculoskeletal and connective tissue diseases are the most common causes of disability, while mental health and behavioral disorders play a relatively small role. Thus, our results suggest that experience rating in DI is an effective policy in reducing the disability inflow at least for worker groups with relatively high disability risks due to musculoskeletal and connective tissue diseases.

Moreover, our results confirm the previous finding that the layoff risk is much higher for older workers who can qualify for extended UI benefits until retirement. We find that large firms in particular exploit this scheme as a soft way to reduce their workforce. We show that coinsurance UI costs mitigate this problem by reducing the excess layoffs of older employees, without having negative spillover effects on the inflow rates to other benefits. This result should be treated with some caution, as the effect is imprecisely estimated and it is not statistically significant in all considered cases.

The lack of adverse spillover effects between DI and UI programs is noteworthy, because it implies that the direct effects of employers' DI and UI costs are not attenuated by higher inflow rates to other benefits. The result is in line with the findings of Kyyrä (2015), who found that stricter eligibility criteria for DI benefits do not increase the unemployment inflow, as well as with the findings of Kyyrä and Pesola (2020), who found that postponing access to the UT scheme by two years did not increase take-up of sickness or disability benefits. But these findings for Finland are at odds with the evidence of substitution between UI and DI programs in Sweden, Austria and the Netherlands in Karlström et al. (2008), Koning and van Vuuren (2010), Lammers et al. (2013), Staubli (2011), Borghans et al. (2014) and Inderbitzin et al. (2016). In Finland, experience-rated firms cannot easily avoid DI costs by laying off their employees with health problems. This is because the DI costs are assigned to the firms in which disability pension recipients worked one and two years prior to the year when the medical condition was first diagnosed, which can be several years before the disability pension was awarded. Moreover, in the case of older workers who can qualify for extended UI benefits, laying off a worker for health problems may be an even less attractive option because the UI costs can be relatively high compared to the expected DI costs.

Declaration of competing interest

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Data availability

The authors do not have permission to share data.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.labeco.2023.102434>. Supplementary material associated with this article can be found, in the online version, at <https://doria.fi/handle/10024/186324>

References

- Anderson, P.M., Meyer, B.D., 2000. The effects of the unemployment insurance payroll tax on wages, employment, claims and denials. *J. Public Econ.* 78 (1–2), 81–106.
- Autor, D.H., 2011. The unsustainable rise of the disability rolls in the United States: Causes, consequences, and policy options. NBER Working Paper 17697, National Bureau of Economic Research.
- Borghans, L., Gielen, A., Luttmer, E.F.P., 2014. Social support substitution and the earnings rebound: Evidence from a regression discontinuity in disability insurance reform. *Am. Econ. J.: Econ. Policy* 6 (4), 34–70.

- Bruce, C.J., Atkins, F.J., 1993. Efficiency effects of premium-setting regimes under workers' compensations: Canada and the United States. *J. Labor Econ.* 11 (1), S38–S69.
- Burkhauser, R., Daly, M.C., 2011. *The Declining Work and Welfare of People with Disabilities: What Went Wrong and a Strategy for Change*. American Enterprise Institute.
- Card, D., Levine, P.B., 1994. Unemployment insurance taxes and the cyclical and seasonal properties of unemployment. *J. Public Econ.* 53, 1–29.
- de Groot, N., Koning, P., 2016. Assessing the effects of disability insurance experience rating. The case of The Netherlands. *Labour Econ.* 41, 304–317.
- de Groot, N., Koning, P., 2022. A burden too big to bear? The effect of experience-rated disability insurance premiums on firm bankruptcies and employment. *Scand. J. Econ.* 124 (1), 214–242.
- Fath, J., Clemens, F., 2005. Experience rating of unemployment insurance in the US: A model for Europe?. CESifo DICE Report, CESifo, pp. 44–50.
- Fath, J., Fuest, C., 2005. Experience rating versus employment protection laws in a model where firms monitor workers. *Scand. J. Econ.* 107 (2), 299–313.
- Hakola, T., Uusitalo, R., 2005. Not so voluntary retirement decisions? Evidence from a pension reform. *J. Public Econ.* 89 (11–12), 2121–2136.
- Hawkins, A., Simola, S., 2018. Paying for disability insurance? Firm cost sharing and its employment consequences. Unpublished manuscript.
- Inderbitzin, L., Staubli, S., Zweimüller, J., 2016. Extended unemployment benefits and early retirement: Program complementarity and program substitution. *Am. Econ. J.* 8 (1), 253–288.
- Johnston, A., 2021. Unemployment insurance taxes and labor demand: Quasi-experimental evidence from administrative data. *Am. Econ. J.: Econ. Policy* 13 (1), 206–293.
- Karlström, A., Palme, M., Svensson, I., 2008. Employment effect of stricter rules for eligibility for DI: Evidence from a natural experiment in Sweden. *J. Public Econ.* 92 (10–11), 2071–2082.
- Koning, P., 2009. Experience rating and the inflow into disability insurance. *De Economist* 157 (3), 315–335.
- Koning, P., Lindeboom, M., 2015. The rise and fall of disability insurance enrollment in the Netherlands. *J. Econ. Perspect.* 29 (2), 151–172.
- Koning, P., van Vuuren, D., 2010. Disability insurance and unemployment insurance as substitute pathways. *Appl. Econ.* 42 (5), 575–588.
- Korkeamäki, O., Kyrrä, T., 2012. Institutional rules, labor demand and retirement through disability programme participation. *J. Popul. Econ.* 25 (2), 439–468.
- Krueger, A.B., 1990. Workers' compensation insurance and the duration of workplace injuries. NBER Working Papers 3253, National Bureau of Economic Research.
- Kyrrä, T., 2015. Early retirement policy in the presence of competing exit pathways: Evidence from pension reforms in Finland. *Economica* 82, 46–78.
- Kyrrä, T., Paukkeri, T., 2018. Does experience rating reduce sickness and disability claims? Evidence from policy kinks. *J. Health Econ.* 61, 178–192.
- Kyrrä, T., Pesola, H., 2020. Long-term effect of extended unemployment benefits for older workers. *Labour Econ.* 62, 101777.
- Kyrrä, T., Tuomala, J., 2013. Does experience rating reduce disability inflow?. VATT Working Papers 46.
- Kyrrä, T., Wilke, R.A., 2007. Reduction in the long-term unemployment of the elderly: A success story from Finland. *J. Eur. Econom. Assoc.* 5 (1), 154–182.
- Lammers, M., Bioemen, H., Hochguertel, S., 2013. Job search requirements for older unemployed: Transitions to employment, early retirement and disability benefits. *Eur. Econ. Rev.* 58 (2), 31–57.
- Ruser, J.W., 1985. Workers' compensation insurance, experience-rating, and occupational injuries. *Rand J. Econ.* 16 (4), 487–503.
- Ruser, J.W., 1991. Workers' compensation and occupational injuries and illness. *J. Labor Econ.* 9 (4), 325–350.
- Staubli, S., 2011. The impact of stricter criteria for disability insurance on labor force participation. *J. Public Econ.* 95 (9–10), 1223–1236.
- Thomason, T., Pozzebon, S., 2002. Determinants of firm workplace health and safety claims management practices. *Ind. Labor Relations Rev.* 55 (2), 286–307.
- Topel, R.H., 1983. On layoffs and unemployment insurance. *Amer. Econ. Rev.* 73 (4), 541–559.
- van Sonsbeek, J.-M., Gradus, R.H., 2013. Estimating the effects of recent disability reforms in the Netherlands. *Oxf. Econ. Pap.* 65 (4), 832–855.
- Woodbury, S.A., Anderson, R., Jacobson, L.S., 2004. Layoffs and experience rating of the unemployment insurance payroll tax: Panel data analysis of employers in three states. Policy report, U.S. Department of Labor.
- Yashiro, N., Kyrrä, T., Hwang, H., Tuomala, J., 2022. Technology, labour market institutions and early retirement. *Econ. Policy* 37 (112), 811–849.